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B.Tech. Degree III Semester Examination November 2016

ME 15-1302 ELECTRICAL TECHNOLOGY (2015 Scheme)

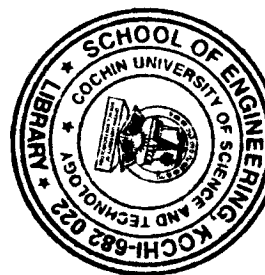
Time : 3 Hours

Maximum Marks : 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) Explain the conditions for voltage build up of a self excited shunt generator.
- (b) Derive an equation for back emf in DC motor.
- (c) Draw and explain the no load phasor diagram of a 1 ϕ transformer.
- (d) Distinguish between power transformer and distribution transformer.
- (e) Obtain the relationship for amount of copper saved in an autotransformer.
- (f) Define voltage regulation of an alternator.
- (g) What is meant by synchronous condenser? Explain
- (h) Derive the condition for maximum running torque of 3 ϕ Induction Motor.
- (i) Define corona. What are the factors affecting corona?
- (j) What are the electrical equipments in power station?



PART B

(4 × 10 = 40)

- II. (a) A 400 V shunt generator has a full load current of 200 A. Its armature resistance is 0.06Ω and the field resistance is 100Ω . The stray losses are 2000 W. Find the hp of the prime mover when it is delivering full load and find the load for which efficiency is maximum. (6)
- (b) Sketch and explain the mechanical and electrical characteristics of DC Shunt motor. (4)

OR

- III. (a) Define commutation. Explain the process of commutation in dc generator with neat sketches. What are the methods for improving commutation? (6)
- (b) Derive the condition for maximum efficiency of a DC generator. (4)

- IV. A 20 kVA, 2000/200 V, 50 Hz, 1 ϕ transformer gave the following results: (10)
 O.C. test (L.V. side) : 200 V, 4 A, 120 W
 S.C. test (H.V. side) : 60 V, 10 A, 300 W.
 (i.) Draw the equivalent circuit referred to HV side.
 (ii.) Calculate the efficiency of the transformer at half load, 0.8 pf lagging.

OR

- V. (a) Develop the exact equivalent circuit of a transformer. From this derive the approximate equivalent circuit as referred to primary. (5)
- (b) A 500 kVA transformer has an efficiency of 95% at full load and also at 60% full load both at upf. Find the core loss and copper loss at full load. Determine the efficiency of the transformer at $\frac{3}{4}$ th full load. (5)

(P.T.O)

- VI (a) Explain the different methods for starting of synchronous motor. (4)
- (b) A three phase, 2 pole, 50 Hz, 2300 V synchronous machine has 42 slots. Each slot has 2 conductors in a double layer winding. The coil pitch is 17 slots. Calculate the flux per pole required to generate a phase voltage of $2300/\sqrt{3}$ V. (6)

OR

- VII A 6 pole , 50 Hz, 3 ϕ induction motor running on full load develops a useful torque of 160 Nm when the rotor emf makes 120 complete cycles per minute. The mechanical torque lost in friction is 10 Nm. The total stator loss is given to be 850 W. Calculate (i) Rotor slip (ii) Efficiency at full load (iii) Rotor Cu loss (iv) Shaft power output. (10)

- VIII With a neat schematic, explain the working of a thermal power plant. (10)

OR

- IX (a) With a neat sketch explain nuclear power station. (6)
- (b) What is skin effect? (4)